Wayne Williams Opanja

Software Engineer and Product Manager · AI, data platforms and digital public services

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PROFILE

Software engineer and product manager with deep expertise in data-collection and analytics platforms, AI products, fintech and digital public services across Africa. Works across the whole product cycle, from user research and journey design to full-stack development, AI agents and automation, QA and deployment. Previously Senior Software Engineer at ProDG Studios, leading frontend and UX on Nigeria's national innovation portal for the Federal Ministry of Education, built to collect data offline in remote areas, and full-stack developer on NGAO, a WhatsApp data-collection platform in Kenya with more than 3,000 users; also delivered the frontend of Venture Garden Group's CrossVest investment marketplace into production. At Pneuma Labs has delivered 8 projects for clients in six countries, met partner CEOs in Uganda and Rwanda, and trained five CEOs and their technical leads in Rwanda. Technical Analyst and Partner at Climate Shapers Africa. Works with TypeScript, Python, Next.js, React, Node.js, NestJS, FastAPI, PostgreSQL, LLMs, WhatsApp and SMS, Docker, Kubernetes and Power BI. CHFI-certified.

EXPERIENCE

Lead Software Engineer · Pneuma Labs | Nairobi, clients in 6 countries | 2026 – Present

AI and automation company: 8 projects for 8 clients in logistics, real estate, mining, financial services and government.

- **PropFlow property-management platform:** designed and built one system for landlords to run rent, leases, repairs and payments, with a WhatsApp line for tenants and emergency phone calls. Travelled to Uganda and Rwanda to meet partners, including the leadership teams of Keystone Technologies and Unispace, and trained five CEOs, property owners and their CTOs in Rwanda to use it. *Next.js, PostgreSQL, Twilio, Paystack, Docker.*
- **Nairobi City County revenue assistant:** turned the county's revenue reports into one clean database and added an AI assistant staff can question on WhatsApp or the web. Checked its answers against the original reports and fixed double-counting before staff relied on it. *Next.js, Supabase/PostgreSQL, LLMs, WhatsApp.*
- **NSE market research assistant:** built a team of AI agents that research Nairobi Securities Exchange stocks every morning, argue the case for and against, and log each call so its accuracy can be measured. Used by analysts in Kenya. *TypeScript, LLMs, Supabase, GitHub Actions.*
- **Process automation:** mapped operations for Mediscope and the Strathmore Centre for Value Investing, designed automations around how their teams work, and trained the Centre's team.

Senior Software Engineer · ProDG Studios | UK, full-time, remote | 2025 – 2026

- **RICC national innovation portal, Federal Ministry of Education, Nigeria:** led frontend and UX on the platform that maps Nigeria's research-to-market pipeline (ricc.education.gov.ng). It works offline in remote areas and syncs when agents are back online. Built the landing page, analytics dashboards, evidence-verification panel and consent controls, and ran end-to-end QA for the national rollout. *Next.js, React, TypeScript, FastAPI, PostgreSQL.*
- **NGAO, WhatsApp data collection for Kenya:** full-stack developer. Built the engines and orchestrators that run the conversations, read answers in English or Kiswahili (even misspelled), take photos, voice and location, and keep every submission as a permanent record, then connected them to the admin dashboards. Rolled out, with more than 3,000 users. *NestJS, PostgreSQL, React, Twilio, WhatsApp, LLMs.*
- **CrossVest, Venture Garden Group:** developed and deployed the frontend of an investment marketplace that partner apps can embed, so people in the diaspora can invest in vetted African opportunities. Took it through user acceptance testing into production, with identity checks through Smile ID and Sumsub.
- **Trademark Registry, Federal Ministry of Industry, Trade and Investment, Nigeria:** built digital registration and administration workflows that replaced manual processes.

CLIENT AND PARTNER ENGAGEMENTS

Consultant, Climate & Nature Toolkit · British International Investment | Remote

- **Made a development-finance framework usable:** built the platform behind BII's climate and nature risk toolkit, so financial institutions can screen clients, measure the emissions linked to their loans and investments, and stress-test portfolios step by step. *Next.js, TypeScript, PostgreSQL.*

Technical Analyst and Partner · Climate Shapers Africa | Nairobi

Joined as a data scientist; promoted to technical analyst; now a partner.

- **Climate risk for banks:** built climate-stress exposure analytics for the Kenya Bankers Association, linking climate indicators to the chance of loan default, with NCBA, Equity Bank and Victoria Commercial Bank among the stakeholders.
- **Kenya Climate & Nature Directory:** built Power BI dashboards and AI research automations for the Directory, and the Africa Climate & Nature Data Platform prototype with a Kenya county risk map.

Penetration Testing Consultant · Harambee SACCO

- Tested a financial cooperative's systems and set out the fixes.

SELECTED PROJECTS

- **MessageHeard v3 (full-stack developer, 2025 – 2026):** built the backend and the interface and trained the custom image models for a multi-tenant AI video platform that breaks a transcript into scenes and generates images and video for each.
- **Chat Your Finances (lead developer):** AI assistant that reads M-Pesa, bank and payslip statements and answers money questions on WhatsApp, Telegram and web. *Next.js, Supabase, OCR.*

EDUCATION AND CERTIFICATIONS

- **Strathmore University:** Bachelor of Business Information Technology (2024 – 2027, in progress, Grade A average); Diploma in Business Information Technology (2023 – 2024). Coursework includes an event-streaming pipeline with Apache Kafka, Docker and PostgreSQL.
- **Strathmore Institute of Mathematical Sciences, Student Assistant (Developer) (2023 – 2025):** built a course evaluation system (Django, React, PostgreSQL) that increased student feedback rates and gave staff timely analytics.
- **Strathmore University ICT, Software Developer Intern (2025 – 2026, degree internship):** rebuilt the phone-billing system as an Odoo module with call-record and HR/ERP sync.
- **Certifications:** CHFI, EC-Council (2024); Applied Data Science & ML, WorldQuant University (2025); Google Cybersecurity; Simplilearn Full-Stack Development.
- **Languages:** English and Swahili (fluent), German (beginner).